MIL-P-46699B (AR)
2 JUNE 1981
SUPERSEDING
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5 August 1975

MILITARY SPECIFICATION

PROPELLANT M1 FOR USE IN CHARGE PROPELLING M2 (WHITE BAG) FOR 8 INCH HOWITZER

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers M1 Propellant for use in Propelling Charge M2 (White Bag) for 8 Inch Howitzers, M2, M2A1, M2A2 and M47.

1.2 Classification. The propellant shall be M1, Cylindrical Multi-Perforated Grain (MP).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

MIL-C-46695 - Charge, Propelling, 8 Inch, M2 (White Bag), Loading, Assembly and Packing

FSC: 1375

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-652 - Propellants, Solid, for Cannons, Requirements and Packing

DRAWINGS (see 6.3)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

8861374 - Charge, Propelling, M2 For 8 Inch Howitzer, Assembly

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procuremnt functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2. Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMC PAMPHLET 706-114

Experimental Statistics, Section 5, Tables

(Application for copies should be addressed to he Publications and Reproduction Agency, Letterkenny Army Depot, Chambersburg, PA 17201).

3. REQUIREMENTS

3.1 Material. The propellant shall be M1, Type 1 selected in accordance with 4.3 and shall conform to the requirements of MIL-STD-652 as determined in 4.4.2.1.

3.2 Ballistic assessment requirements. The propellant when fired in the 8 Inch Howitzer, M2 and with the weight of projectile HE M106 (200.0 lbs) shall comply with the following requirements at 70 degrees Fahrenheit (^{o}F) and standard weapon conditions (see 4.5.1). The propellant at zone 7 shall be capable of assessment to a velocity of 1,950 feet per second (fps) with an average pressure of 39,600 pounds per square inch (psi) maximum.

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3.3 Ballistic uniformity requirements. The propellant when fired in the 8" Howitzer M2 with the projectile HE M106 (see 4.5.1) shall comply with the following requirements when tested as specified in 4.5.1.2. All charges (reference, master calibration and test) shall be manufactured in acordance with MIL-C-46695.

3.3.1 Uniformity of velocity. The standard deviation of velocity for the lot at 70°F shall not exceed 5.5 fps (see 4.4.2.4) at zone 7.

3.3.2 Average pressure. At 70°F and standard weapon conditions, the propellant shall not produce an average pressure greater than 39,600 psi at zone 7.

3.3.3 Individual maximum pressure (IMP). (see 6.6) The propellant shall not produce an individual pressure greater than 45,540 psi at any operating temperature.

3.4 Initial production. This specification contains provisions for initial production inspection. Requirements for the submission of initial production inspection samples by the contractor shall be as specified in 4.5.1.2.1.

3.5 Preproduction inspection. This specification contains provisions for preproduction inspection. Requirements for the submission of preproduction inspection samples by the contractor shall be as specified in 4.3.1.

3.6 Workmanship. The best commercial practices shall be used in the formulation of propellant furnished under this specification, and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes. Unless otherwise specified, no reground propellant or nitrocellulose shall be used in the manufacture of propellant procured under this specification.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classifiecation of inspection. The following types of inspection shall be conducted on the unit product:

a. Preproduction Sample Inspection
b. Quality Conformance Inspection

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4.3 Preproduction sample (Pilot lots). The preproduction sample is intended to provide the necessary information to establish the propellant granulation needed to meet the required ballistic parameters.

4.3.1 Submission. The Contractor shall select three (3) propellant granulations which, from experience, should best fit the requirements of this specification. The contractor shall submit a preproduction sample as designated by the Contracting Officer for evaluation in accordance with 4.3.2, consisting of three (3) pilot lots of 1500 lbs to 1800 lbs each representing the three (3) candidate granulations.

4.3.2 Inspections to be performed. Each of the pilot lots comprising the preproduction sample and submitted in accordance with 4.3.1 may be subjected by the Government to any or all of the requirements of this specification.

4.3.3 Rejection. Government Engineering and Product Assurance personnel and the Contractor shall review the pilot lot test data and shall select a granulation size which best fits the requirements of this specification. The Technical Agency shall notify the Contracting Officer of the selected granulation. The Contractor shall request approval from the Contracting Officer to produce the selected granulation. If none of the pilot lots submitted meets the requirements of this specification, a new preproduction sample shall be submitted in accordance with 4.3.1. All future production lots should be of the same basic granualtion.

4.4 Quality Conformance Inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination and testing.

PRECAUTION - This specification covers sampling and testing of chemical, toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.4.2.1 Sampling for laboratory testing (see 3.1). Each lot of propellant shall be sampled and tested for composition, form, dimensions and chemical/physical properties in accordance with MIL-STD-652. (see 6.4 and 6.5).

4.4.2.2 Sampling for ballistic testing (see 3.2). Each lot of propellant shall be sampled for Proving Ground tests in accordance with MIL-STD-652.

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4.4.2.3 Charge establishment (see 3.2). If the lot does not meet the applicable requirements, testing shall cease and the Test Director shall immediately notify the procuring activity.

4.4.2.4 Velocity uniformity (70 degrees F) (see 3.3.1). If the product of the factor (see 6.8) times the velocity standard deviation of the rounds tested in accordance with 4.5.1.2 exceeds the applicable requirement, apply referral criteria in accordance with 4.5.1.4. If the referral criteria cited do not apply, the lot shall be rejected.

4.4.2.5 Average pressure (see 3.3.2). If the average pressure obtained exceeds the applicable requirements, apply referral criteria in accordance with 4.5.1.4. If the referral criteria cited do not apply, the lot shall be rejected.

4.4.2.6 Individual maximum pressure (IMP) (see 3.3.3).

4.4.2.6.1 Initial production. If any individual pressure observed during uniformity firings at any operating temperature after correction for difference between as fired and recommended service charge weight exceeds the applicable requirement the lot shall be rejected.

4.4.2.6.2 Subsequent production. If any individual pressure observed during the 70°F uniformity series when corrected to standard conditions and recommended charge weight exceeds 45,540 psi minus the quantity 75 times the pressure/temperature coefficient established during Initial Production Testing, the lot shall be tested at +145°F in accordance with 4.5.1.2.1. If any individual pressure at 145°F when corrected to standard conditions and recommended charge weight exceeds 45,450 psi the lot shall be rejected.

4.5 Test methods and procedures

4.5.1 Proving ground assessments (See Table I). This test shall be performed at a Government Proving Ground in accordance with the applicable acceptance test procedure using M106, HE projectiles inert loaded within 0.20 lb of an as fired weight of 200.0 lb using Primer, M82 and fired in a 8" Cannon, M2 series. All charges shall be temperature conditioned in a large conditioning box, with adequate air circulation to assure uniform conditioning, at the required temperature for a minimum of 24 hours and assurance will be made that, when

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fired, the charges are at the conditioning temperature. The following tolerances shall apply to all temperature conditionings: -60°F + 5°F; 70°F, ± 2.5°F; 145°F, -5°F. Copper crusher gages (2) shall be used to measure the maximum chamber pressure of each charge. Velocity coils will be used to measure the muzzle velocity of each charge. Test charges shall be assembled in accordance with drawing 8861374 and MIL-C-46695.

4.5.1.1 Charge establishment

4.5.1.1.1 Charge slope method. Three or more groups of charges loaded with selected charge weights of propellant (weights selected should bracket expected service charge weight) shall be conditioned and fired at 70°F. Calibration charges at each zone shall also be conditioned and fired at 70°F with the test charges. From the data obtained, a charge weight shall be calculated which will yield the prescribed service velocity (see 3.2.). This calculated charge weight shall then be loaded into charges for the uniformity series (see 4.5.1.2.1). The difference between the corrected velocity obtained during the uniformity series and the service velocity shall be compensated for by a final adjustment in charge weight using the slope of the velocity charge weight curve previously established. The resultant will be the recommended charge weight.

4.5.1.1.2 Constant slope method. After ten (10) lots of a given production series have been fired to provide a good knowledge of the propellant characteristics, the charge slope method of assessment may be discontinued and the charge weight determined by the constant slope method. A composite charge weight versus velocity curve and charge weight versus pressure curve can be established from test results generated. This curve can then be used as in 4.5.1.1.1 to calculate the charge weight needed to achieve service velocity. This calculated charge weight is then to be loaded into charges for the uniformity series and final adjustment is to be made as indicated above for the charge slope method (see 4.5.1.1.1). The resultant will be the recommended charge weight.

4.5.1.2 Uniformity

4.5.1.2.1 Initial Production. Thirty rounds shall be assembled with the calculated charge weight of propellant. Ten rounds each shall be conditioned to -60°F, 70°F and 145°F

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(as indicated by thermocoupled control round). Rounds fired at 70°F shall be fired alternately with calibration rounds and shall be corrected to standard weapon/firing conditions (i.e. the difference between calibration performance under test conditions and standard conditions). Firing results at all temperatures shall be corrected to standard projectile weight (see 4.5.1) and to recommended charge weight conditions (i.e. the difference between as fired charge weight and recommended charge weight). The means and standard deviation of the velocities and pressures at all firing temperatures shall be calculated using corrected results. The average pressure-temperature coefficient for the lot shall be calculated from the pressure data obtained at 70°F and 145°F. The average of the coefficients for ten lots shall be utilized to calculate a 70°F upper pressure control for use in subsequent production. see (6.7)

4.5.1.2.2 Subsequent production. Ten (10) charges shall be assembled with the calculated charge weight (see 4.5.1.1.2) and conditioned at 70°F (as indicated by thermocoupled control charge). Charges fired at 70°F shall be fired alternately with calibration charges and shall be corrected to standard weapon/firing conditions (i.e. the difference between calibration performance under test conditions and standard conditions). Firing results shall be corrected to standard projectile weight (see 4.5.1) and to recommended charge weight conditions (i.e., the difference between as fired charge weight and recommended charge weight). The means and standard deviations of the velocities and pressures shall be calculated using corrected results. Throughout production, if 3 consecutive lots fail on first test, testing shall revert back to the initial production plan (see 4.5.1.2.1). If any corrected individual pressure exceeds the calculated 70°F upper pressure control (see 4.4.2.6.2) ten additional test rounds shall be conditioned and fired at 145°F. All results at 145°F shall be corrected to standard projectile weight (see 3.2) and recommended charge weight conditions. If any individual round at 145°F exceeds 45,540 psi, the lot shall be rejected.

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4.5.1.3 Retest criteria. A lot shall be retested whenever any of the following conditions exist or whenever the proving ground deems it necessary.

a. The final adjustment in charge weight exceeeds two percent of the charge weight used in uniformity firings, retest for uniformity in accordance with 4.5.1.2.

b. If for any reason, the Proving Ground considers that test conditions have detrimentally affected the test results, additional charges as required shall be tested.

4.5.1.4 Referral criteria. A test lot shall be referred to the procuring activity for dispostion whenever any of the following conditions exist or whenever the proving ground deems it necessary.

a. Calibration rounds fired in any test phase fail to comply with Section 3 of this specification.

b. Calibration correction applied exceeds two percent of the expected muzzle velocity.

c. Calibration components are not available.

d. Number of test or calibration results utilized in calculations is less than specified.

e. Any unusual occurrences during ballistiac testing, such as excessive delays between rounds, stickers, poor projectile flights, etc. Each such event shall be reported in detail on the applicable firing record.

f. Test lot fails velocity standard deviation or average pressure requirement at 70°F and any extreme test value is an outlier. The following test shall be performed to determine if a single test result is an outlier. With a sample size of 10 at a significance level of 0.05, the critical value (T_c) is 0.477. Calculate a test value (T_t) as follows:

For Low Outlier

$$T_t = \frac{(X_2 - X_1)}{(X_9 - X_1)}$$

For High Outlier

$$T_t = \frac{(X_{10} - X_9)}{(X_{10} - X_2)}$$

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Where:

X_1 = test result being tested as low outlier
X_2 = test result adjacent to X_1, after all the test results are ordered from smallest to largest
X_9 = test result adjacent to the largest test result
X_{10} = test result being tested as high outlier

X_1 or X_{10} is an outlier if T_t is greater than T_C.

5. PACKAGING

5.1 Packing and marking. Packing and marking shall be in accordance with MIL-STD-652, except that all propellant designated for use as reference shall be packed level A (see 6.5).

6. NOTES

6.1 Ordering data. Ordering data shall be in accordance with MIL-A-48078 with the following exceptions:

6.1.1 Procurement requirements. Procurement requirements of MIL-A-48078 shall apply.

6.1.2 Contract data requirements. One copy each of the Propellant Description Sheet and the Acceptance Test Summary shall be forwarded to each office designated by the contracting officer. Copies of data item descriptions required by the contractor in connection with specific procurement functions should be obtained from the procuring activity.

6.2 Proving ground test summary (see Table I).

6.3 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the custodianship of ARRADCOM.

6.4 Surveillance sample. Samples shall be forwarded to US Army Armament Research and Development Command, Dover, NJ 07801, ATTN: DRDAR-LCE-P for the propellant surveillance test program in accordance with MIL-STD-652.

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TABLE I

PROVING GROUND TEST SUMMARY

Test Phase	Temperature (°F)	Sample Size Cal	Sample Size Test	Velocity Requirements Service Vel (fps)	Velocity Requirements Std Dev (fps)	Pressure Requirements IMP (psi)	Pressure Requirements AVG Pressure (psi)
Uniformity							
Initial	-60	NA	10	NA	NA	45,540	NA
Production	70	10	10	1950	5.5	45,540	39,600
	145	NA	10	NA	NA	45,540	NA
Subsequent Production	70	10	10	1950	5.5	see 4.5.1.2.2	39,600
	145*	NA	10	NA	NA	45,540	NA

*To be fired only when 70°F results indicate that marginal pressure performance may exist (see 4.5.1.2.2)

NA-not applicable

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6.5 Reference sample. Whenever a propellant lot is selected for reference use, a 5-lb sample shall be forwarded to US Army Armament Research and Development Command, Dover, NJ 07801, ATTN: DRDAR-LCE-P for use in evaluating the propellant stockpile.

6.6 Individual Maximum Pressure. Individual maximum pressure specified in 3.3.3 is based on firing with an M106 projectile loaded to an as fired weight of 200.0 lbs. Any change in weight or projectile will require an adjustment of this pressure requirement.

6.7 Initial production lot. The initial production lots represents the first ten consecutive lots from each manufacturer which have been produced and met the acceptance criteria of this specification. Lapse in production of 90 days or more shall cause testing to revert back to initial production plan.

6.8 Factor. The factor, as taken from Table A-21 of AMC Pamphlet 706-114 using the 95 percent upper confidence limits (A.05) and n-1 degrees of freedom, makes allowance for the probability that a standard deviation from a smaller sample size exceeds the true standard deviation by chance alone. The standard deviation shall be calculated with (n-1) as the divisor in a standard statistical technique equivalent to that shown in MIL-A-48078.

Custodian:
Army-AR

Preparing Activity:
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